



Understanding HDF5 in SNP-Seek

What it is, why we use it, and how genotype data gets there

What is HDF5?

Hierarchical Data Format version 5



Binary Container

Stores massive datasets in a single file — like a filesystem within a file



Fast Random Access

Read any slice of data without loading the whole file into memory



Chunked Storage

Data is divided into chunks for efficient compression and retrieval



Self-Describing

Contains metadata about dimensions, types, and structure

Think of it like...

A spreadsheet with
5 million+ rows
and 4,000+ columns

...but stored so you
can instantly jump to
any cell without
opening the whole file.

HDF5 by the Numbers

Real example: RICE-RP dataset in SNP-Seek

5,231,433

SNP positions

4,591

rice samples

~24 Billion

genotype data points

< 1 sec

query response time

Why HDF5 over Text Files?

	Text Matrix	HDF5
File Size	Very large (uncompressed)	Compressed (much smaller)
Read Speed	Must scan entire file	Jump to any location
Partial Query	Not possible	Read only what you need
Human Readable	Yes (good for checking)	No (binary format)
Web Portal Use	Too slow for queries	Powers SNP-Seek



Key Insight

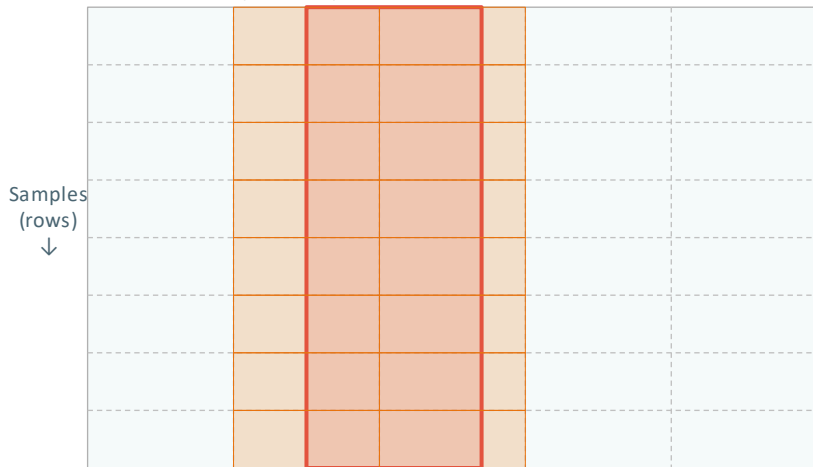
The text matrix is the human-friendly checkpoint — you can open it and verify the data. HDF5 is the machine-optimized version that makes SNP-Seek fast enough for a web browser.

How HDF5 chunking works

Chunk shape determines query speed — HDF5 always reads whole chunks

Current: 20 samples per chunk

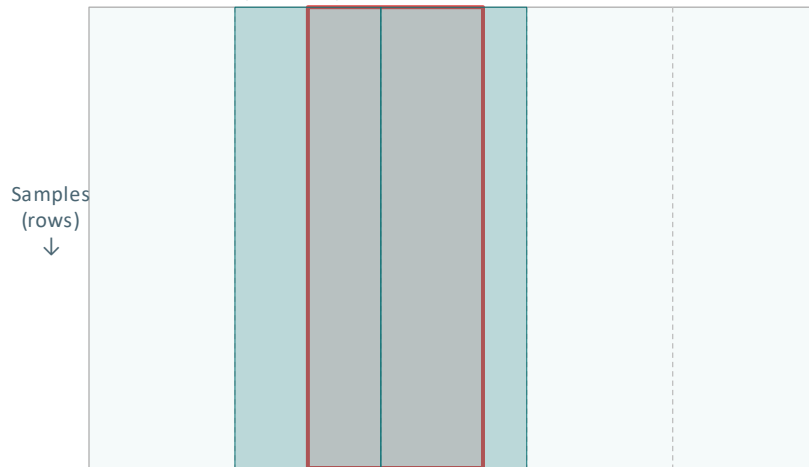
-m 5231 -n 20
SNPs (columns) →



Query region Chunks loaded Chunk boundaries

Ideal: all samples per chunk

-m 5231 -n 4591
SNPs (columns) →



HDF5 reads whole chunks. A B-tree index maps each chunk to its byte position on disk — no scanning.

The Loading Pipeline

How raw genotype files become an HDF5 file for SNP-Seek



What Goes In: VCF & PLINK

The two standard genotype file formats

VCF

Variant Call Format

- Merged multisample file
- SNPs only (no indels)
- Contains genotype calls per sample
- Most common sequencing output

PLINK

Binary PED format (.bed/.bim/.fam)

- Binary genotype data + map
- SNPs only (no structural variants)
- Needs REF allele verification
- Common in GWAS studies

 **Important: Only SNP genotypes — indels must be processed separately**

The Text Matrix

The human-readable intermediate step

CHR	POS	REF	ALT	Sample1	Sample2	Sample3	...
1	1151	C	A	CC	CC	CC	...
1	1173	C	T	CC	CC	CC	...
1	1178	G	T	TT	GG	TT	...
1	1203	T	C	CC	TT	CC	...
1	1248	G	A	00	GG	GG	...
1	1249	A	C	CC	AA	CC	...

Rows = SNPs

Each row is one SNP position across the genome

Columns = Samples

Diploid genotype calls
e.g. CC, AG, TT

Missing Data

Encoded as 00 or ..
(shown in red above)

Three Output Files

Generated from the text matrix conversion step



HDF5 File

The genotype matrix in binary format — optimized for random access queries

e.g. `ricerp.h5`



Sample IDs

List of sample names in the same order as columns in the HDF5 file

e.g. `sample_list.txt`



SNP Positions

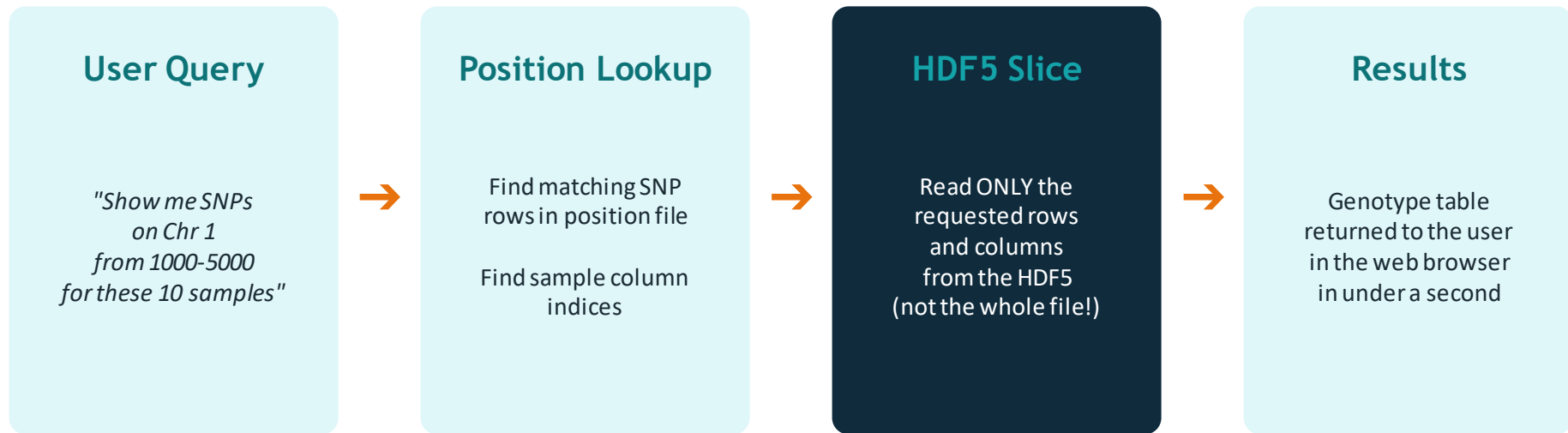
CHR, POS, REF, ALT for each SNP — same row order as HDF5

e.g. `pos.txt`

! Order matters — all three files must be in the same order (same SNP rows, same sample columns)

How SNP-Seek Uses HDF5

The HDF5 file is the engine behind SNP-Seek's genotype queries



```
$ h5ls ricerp.h5
Dataset {5231433, 4591}    # SNPs × Samples
```

Key Takeaways

- ✓ HDF5 is what makes SNP-Seek fast — random access to billions of data points
- ✓ The text matrix is the human-readable checkpoint for verification
- ✓ VCF/PLINK files are converted to a text matrix as an intermediate step
- ✓ Three files work together: HDF5 + Sample IDs + SNP Positions
- ✓ Order consistency across all three files is critical

Next: Hands-on with the conversion commands + Loading into SNP-Seek